3D Reality Capture & BIM — Foundations

The domain, not the project: how 3D measurement, point-cloud learning and building information fit together — plus things to read and follow casually.

General-fluency primer · complements (does not repeat) the project reading list · Oct 2026 · every link verified against primary sources

> **What this is.** The project reading list covers the specific pipeline papers (registration, IFC tooling, BCF, LOA, carbon databases). This one is for the field itself: the conceptual spine of 3D geometry, measurement and building information. Nothing here duplicates that list. §1–3 are the ideas worth genuinely internalising; §4–6 are the modern landscape; §7–8 are for relaxed, ongoing absorption.

If you only do three things: (1) read Solà's **micro Lie theory** (§1) and Lague's **M3C2** (§3), so you think in poses with uncertainty and in significant, not raw, differences; (2) skim **Volk et al. on BIM for existing buildings** (§5) and the **RICS whole-life carbon standard** (§6) to speak the field's language; (3) follow one or two idle sources from §8 (start with **Stachniss on YouTube**) so the field stays in your peripheral vision.

1 — Why 3D Is Different (geometry you can't hand-wave)

A micro Lie theory for state estimation in robotics — Solà, Deray & Atchuthan (2018), arXiv
The clearest intuitive route into SO(3)/SE(3), exp/log maps and perturbations on manifolds, which is what every registration optimiser is secretly doing.
arxiv.org/abs/1812.01537

Lie Groups for 2D and 3D Transformations — Ethan Eade (rev. 2017), self-published note
Short, derivation-first companion to Solà; the reference you keep open while implementing pose updates.
ethaneade.com/lie.pdf

Visualizing quaternions — Grant Sanderson & Ben Eater (2018), interactive explorable
Builds the 4D intuition for unit quaternions and why they avoid gimbal lock, without any algebra up front.
eater.net/quaternions

On the Continuity of Rotation Representations in Neural Networks — Zhou, Barnes, Lu, Yang & Li (2019), CVPR
Why quaternions and Euler angles are bad regression targets and 6D representations are good: geometry meeting deep learning.
arxiv.org/abs/1812.07035

Associating Uncertainty With Three-Dimensional Poses for Use in Estimation Problems — Barfoot & Furgale (2014), IEEE T-RO 30(3) paywalled
How to put a covariance on a rotation and a pose, and propagate it. Registration without error bars is only half an answer.
doi.org/10.1109/TRO.2014.2298059

2 — Point-Cloud Processing Ideas (what you do before any model sees the data)

Surface reconstruction from unorganized points — Hoppe, DeRose, Duchamp, McDonald & Stuetzle (1992), ACM SIGGRAPH Computer Graphics 26(2) paywalled
The origin of PCA-based normal estimation and consistent orientation, the quiet workhorse behind most point-cloud geometry.
doi.org/10.1145/142920.134011

Multidimensional binary search trees used for associative searching — Jon Louis Bentley (1975), Communications of the ACM 18(9) paywalled
The k-d tree paper. Every neighbourhood query, normal and ICP correspondence step leans on this structure.
doi.org/10.1145/361002.361007

Semantic point cloud interpretation based on optimal neighborhoods, relevant features and efficient classifiers — Weinmann, Jutzi, Hinz & Mallet (2015), ISPRS J. Photogrammetry & Remote Sensing 105 paywalled
Shows that the choice of neighbourhood size and shape changes everything downstream; the classic argument for multi-scale thinking.
doi.org/10.1016/j.isprsjprs.2015.01.016

Graduated Non-Convexity for Robust Spatial Perception — Yang, Antonante, Tzoumas & Carlone (2020), IEEE Robotics and Automation Letters 5(2)
A clean unifying view of robust estimation (outliers, M-estimators, why RANSAC is one answer among several) with a practical alternative.
arxiv.org/abs/1909.08605

MAGSAC++, a fast, reliable and accurate robust estimator — Barath, Noskova, Ivashechkin & Matas (2020), CVPR
What modern RANSAC descendants look like: no hand-tuned inlier threshold. Useful for seeing how the idea has matured.
arxiv.org/abs/1912.05909

Past, Present, and Future of Simultaneous Localization And Mapping — Cadena et al. (2016), IEEE T-RO 32(6)
The field's own map of itself: front-end versus back-end, outliers, loop closure, and what robust perception should mean.
arxiv.org/abs/1606.05830

3 — Measurement & Metrology Thinking (what a number from a scanner actually means)

Evaluation of measurement data: Guide to the expression of uncertainty in measurement (GUM) — JCGM 100:2008, BIPM
The international standard vocabulary and method for uncertainty budgets and error propagation. Dense but definitive.
bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf

Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results (TN 1297) — Taylor & Kuyatt (1994), NIST
A short, readable distillation of the GUM; the quickest way to get type A/B uncertainty and combined standard uncertainty into your head.
physics.nist.gov/Pubs/guidelines/TN1297/tn1297s.pdf

Statistics notes: Absence of evidence is not evidence of absence — Altman & Bland (1995), BMJ 311
Two pages on the logic behind "not observed does not mean not there". Reread it every time a scan has a hole where a wall should be.
doi.org/10.1136/bmj.311.7003.485

Accurate 3D comparison of complex topography with terrestrial laser scanner (M3C2) — Lague, Brodu & Leroux (2013), ISPRS J. Photogrammetry & Remote Sensing 82 paywalled
Cloud-to-cloud comparison with a per-point confidence interval, so you can say whether a change is significant. The template for honest deviation maps.
doi.org/10.1016/j.isprsjprs.2013.04.009

4 — Deep Learning on 3D (from PointNet to foundation models)

Deep Learning for 3D Point Clouds: A Survey — Guo, Wang, Bennamoun, Lu, Huang & Wan (2021), IEEE TPAMI 43(12)
The standard survey of classification, detection and segmentation architectures up to 2020; read it for the taxonomy.
arxiv.org/abs/1912.12033

PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation — Qi, Su, Mo & Guibas (2017), CVPR
The founding idea: respect permutation invariance and consume raw points directly.
arxiv.org/abs/1612.00593

PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space — Qi, Yi, Su & Guibas (2017), NeurIPS
Adds local neighbourhoods and hierarchy, the pattern (sample, group, aggregate) that most later models inherit.
arxiv.org/abs/1706.02413

Dynamic Graph CNN for Learning on Point Clouds — Wang, Sun, Liu, Sarma, Bronstein & Solomon (2019), ACM Trans. Graphics 38(5)
EdgeConv on a feature-space k-NN graph: point clouds as dynamic graphs.
arxiv.org/abs/1801.07829

KPConv: Flexible and Deformable Convolution for Point Clouds — Thomas, Qi, Deschaud, Marcotegui, Goulette & Guibas (2019), ICCV
A true convolution with kernel points in continuous space; strong on outdoor and building-scale segmentation.
arxiv.org/abs/1904.08889

Point Transformer — Zhao, Jiang, Jia, Torr & Koltun (2021), ICCV
Vector self-attention over local neighbourhoods; the start of the transformer era for point clouds.
arxiv.org/abs/2012.09164

Point Transformer V2: Grouped Vector Attention and Partition-based Pooling — Wu, Lao, Jiang, Liu & Zhao (2022), NeurIPS
Fixes the efficiency and expressiveness issues of v1; the middle step of the series.
arxiv.org/abs/2210.05666

Point Transformer V3: Simpler, Faster, Stronger — Wu et al. (2024), CVPR
Trades exact neighbour search for serialised point order and scales up cleanly. The current default backbone.
arxiv.org/abs/2312.10035

Pointcept — Pointcept contributors, GitHub
The codebase behind PTv1-v3 and Sonata, with unified benchmarks on ScanNet, S3DIS and others. Read the configs to see how the field trains.
github.com/Pointcept/Pointcept

VGGT: Visual Geometry Grounded Transformer — Wang, Chen, Karaev, Vedaldi, Rupprecht & Novotny (2025), CVPR
Feed-forward prediction of cameras, depth and point maps from images: the learned replacement for classical structure-from-motion.
arxiv.org/abs/2503.11651

Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges — Bronstein, Bruna, Cohen & Veličković (2021), arXiv
The symmetry-first framing (equivariance to SE(3), permutation) that explains why 3D architectures look the way they do.
arxiv.org/abs/2104.13478

5 — BIM, Digital Twins & Information Modelling (the field behind the file format)

Building Information Modeling: Why? What? How? — Borrmann, König, Koch & Beetz, eds. (2018), Ch. 1, Springer, open access
A compact conceptual introduction to BIM, openBIM and model-based collaboration, written for engineers.
doi.org/10.1007/978-3-319-92862-3_1

Building Information Modeling (BIM) for existing buildings: Literature review and future needs — Volk, Stengel & Schultmann (2014), Automation in Construction 38 paywalled
Why as-built BIM is hard: the canonical statement of the problem this whole project lives in.
doi.org/10.1016/j.autcon.2013.10.023

Semantic web technologies in AEC industry: A literature overview — Pauwels, Zhang & Lee (2017), Automation in Construction 73 paywalled
Why IFC alone is not enough and what linked data adds; the information-science side of BIM.
doi.org/10.1016/j.autcon.2016.10.003

BIMForum Level of Development (LOD) Specification — BIMForum (2025), BIMForum
The industry reference for how much a model element is allowed to be trusted at each stage. Free, CC-licensed.
bimforum.org/LOD/

Digital Twin: Mitigating Unpredictable, Undesirable Emergent Behavior in Complex Systems — Grieves & Vickers (2016), Transdisciplinary Perspectives on Complex Systems, Springer paywalled
The widely cited origin of the term. Read it to see how much of today's usage stretches the original meaning.
doi.org/10.1007/978-3-319-38756-7_4

Digital twins — Michael Batty (2018), Environment and Planning B 45(5) paywalled
A short, sceptical editorial on what a digital twin is and is not: a useful counterweight to vendor definitions.
doi.org/10.1177/2399808318796416

6 — Sustainability of the Built Environment (carbon, materials and reuse, conceptually)

Biogenic carbon in buildings: a critical overview of LCA methods — Hoxha et al. (2020), Buildings and Cities 1(1)
A clear tour of why timber and bio-based carbon accounting is contentious: a good lesson in method choices driving results.
doi.org/10.5334/bc.46

Concept for a BIM-based Material Passport for buildings — Honic, Kovacic & Rechberger (2019), IOP Conf. Series: Earth and Environmental Science 225
What a material passport is and how it would hang off a BIM model; the bridge between this project and reuse.
doi.org/10.1088/1755-1315/225/1/012073

Whole life carbon assessment for the built environment, 2nd edition — RICS (2023), RICS professional standard
The professional standard that defines how whole-life carbon is scoped and reported, in force since July 2024.
rics.org/profession-standards/rics-standards-and-guidance/sector-standards/building-surveying-standards/whole-life-carbon-assessment-2nd-edition

Embodied Carbon Primer — LETI (2020), London Energy Transformation Initiative
A practitioner-friendly primer on embodied carbon across life-cycle stages, with design levers. Free.
leti.uk/ecp

Global Status Report for Buildings and Construction 2025–2026 — UNEP & GlobalABC (2026), UNEP
The macro view: where buildings sit in global emissions and what policy is doing about it.
unep.org/resources/report/global-status-report-buildings-and-construction-2025-2026

7 — Books & Longer Reads (mostly free / open)

State Estimation for Robotics (2nd ed.) — Timothy Barfoot (2024), Cambridge University Press; free PDF from the author
Rigid-body kinematics, Lie groups, batch least squares and uncertainty, written for the exact maths this project uses.
asrl.utias.utoronto.ca/~tdb/bib/barfoot_ser24.pdf

Computer Vision: Algorithms and Applications (2nd ed.) — Richard Szeliski (2022), Springer; free PDF via form
The broad 3D vision reference: cameras, alignment, SfM, depth, neural rendering.
szeliski.org/Book/

Modern Robotics: Mechanics, Planning, and Control — Lynch & Park (2017), Cambridge University Press; free preprint
Chapters 2-3 are the gentlest rigorous treatment of rotations and SE(3) available,.
hades.mech.northwestern.edu/index.php/Modern_Robotics

Multiple View Geometry in Computer Vision (2nd ed.) — Hartley & Zisserman (2004), Cambridge University Press paywalled
The classic on projective geometry and estimation; worth owning for the chapters on estimation and error analysis.
robots.ox.ac.uk/~vgg/hzbook/

Airborne and Terrestrial Laser Scanning — Vosselman & Maas, eds. (2010), Whittles Publishing paywalled
The standard text on how laser scanners work and how their data is processed; ISPRS Karl Kraus Medal winner.
whittlespublishing.com/Airborne_and_Terrestrial_Laser_Scanning

BIM Handbook: A Guide to Building Information Modeling (3rd ed.) — Sacks, Eastman, Lee & Teicholz (2018), Wiley paywalled
The field's standard handbook: what BIM is, who uses it and how workflows change.
wiley.com/en-us/BIM+Handbook (ISBN 978-1-119-28757-5)

Sustainable Materials: With Both Eyes Open — Allwood, Cullen et al. (2012), UIT Cambridge; free online
Material efficiency as a climate lever: why using less steel and cement beats making it greener. Free chapters.
withbotheyesopen.com

8 — Idle Learning (follow casually: active as of Oct 2026)

Cyrill Stachniss (YouTube) — Cyrill Stachniss, University of Bonn, YouTube
Full lecture series on photogrammetry and mobile sensing plus the short "5 Minutes with Cyrill" explainers. Uploads in Oct 2026.
youtube.com/@CyrillStachniss

Radiance Fields — Radiance Fields, news site + newsletter
Independent daily coverage of Gaussian splatting, NeRFs and capture tools. The best single feed for fast-moving reality capture.
radiancefields.com

Geo Week News — Diversified Communications, news site
Lidar, reality capture, BIM, digital twins and surveying industry news.
geoweeknews.com

AEC Magazine — X3DMedia, magazine + site
AEC technology reporting: BIM, AI, reality capture, sustainability. Print and web.
aecmag.com

OSArch — OSArch community, community news + forum
Hub of the open-source BIM community: Bonsai, IfcOpenShell, IFC tooling, including point-cloud support work.
osarch.org

Building Tomorrow: Conversations with Climate Solvers — Urban Green Council, podcast
Interviews on building decarbonisation, policy and retrofit; episodes through mid-2026.
urbangreencouncil.org/resource-library/insights/building-tomorrow-conversations-with-climate-solvers/

Carbon Leadership Forum — Carbon Leadership Forum, blog + resources
Embodied-carbon research, policy and tools from the field's main non-profit; posting through 2026.
carbonleadershipforum.org

Every link verified to resolve against a primary source (Oct 2026). Journal and book items marked paywalled (Elsevier, ACM, IEEE, Sage, Springer, Wiley, CUP) block automated fetchers or sit behind subscriptions; they were confirmed real through DOI resolution and Crossref metadata. The UNEP report page blocks automated fetchers too and was confirmed by search. Idle-learning sources were checked for 2026 activity. Dropped as unconfirmed: the Talking Papers podcast (no 2026 episodes found), Computer Vision News (publisher exited in 2024), Thornton Tomasetti's embodied-carbon podcast and Building Transparency's blog (site blocked, 2026 activity not confirmed). Seven items that also appear in the project reading list were removed here to avoid repetition. Companion document: the *Scan vs BIM* project reading list.